Automated Michelson Interferometer

PHYS 491 Lab Report

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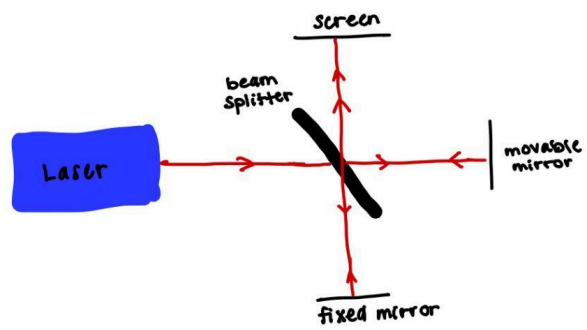
Michelson interferometers are one of the most fundamental tools in experimental physics and optical measurement. Developed by Albert A. Michelson in the late 19th century, these devices showcase how light interference is used to measure refractive indices, wavelengths, extremely small distances, and physical changes with precision.

Their importance is used in many different ways, including:

- Tiny surface features of materials
- Laser wavelength calibration
- Refractive index determination
- Spectroscopy
- Detection of gravitational waves (e.g., LIGO)



The Michelson interferometer operates on the principle of splitting a light beam into two paths using beam splitter. The two beams reflect off separate mirrors and recombine at the beam splitter. The resulting interference pattern depends on the difference in optical path lengths of the two beams. Additionally, small changes in the mirror' position create shifts in the interference fringes, allowing highly sensitive displacement measurements. Interference occurs when two beams travel distances such that their peaks and troughs line up, producing bright and dark fringes.

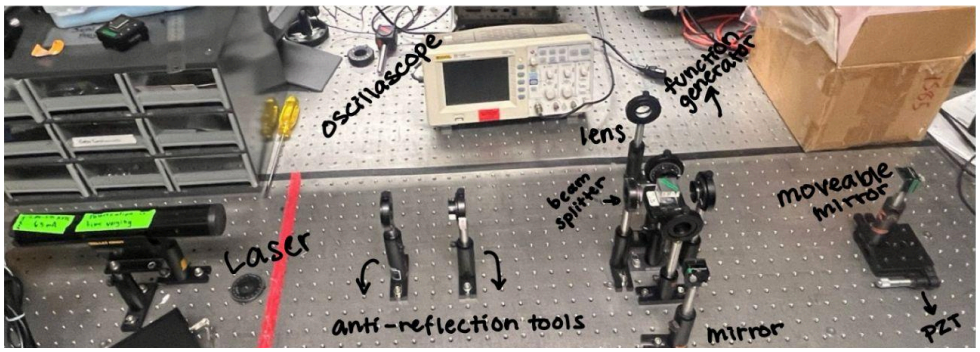
Michelson interferometers are widely used in fundamental physics, material science, engineering, astronomy, and meteorology. They can do surface profiling and deformation analysis, thin film thickness measurements, and stellar diameter measurements.

I. Setup

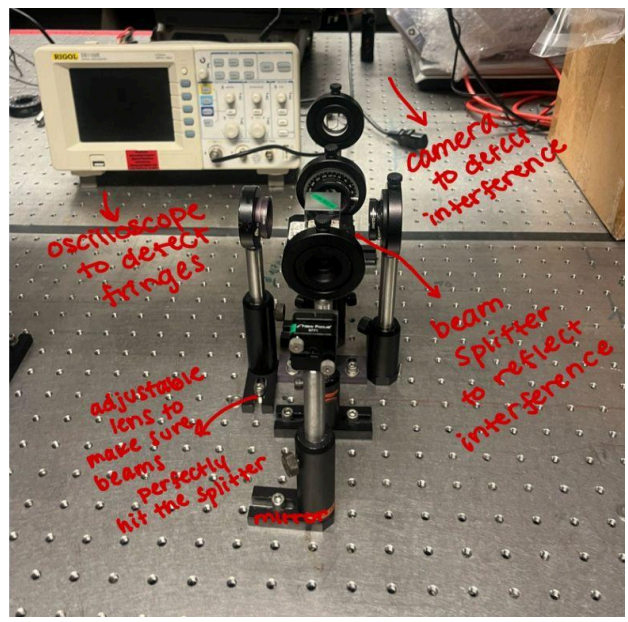
The primary components used in the setup included:

- **Laser Source:** Helium-Neon laser with 6.5mA
- **Beam Splitter:** A 50/50 partially reflective beam splitter is placed at 45° to divide the laser beam into two perpendicular paths.
- **Mirrors:** Two reflective mounted mirrors for precise angular adjustment. One mirror was fixed, and the other was attached to a micrometer stage and PZT actuator for controlled displacement.
- **PZT:** The piezoelectric transducer (PZT) allowed for fine displacement adjustments on the nanometer scale.

- **Micrometer Stage:** Provided larger displacement adjustments with a mechanical screw and thimble.
- **Optical Components:** Additional lenses and anti-reflection tools were placed in the laser beam path to enhance beam shape and minimize stray reflections.
- **Photodetector & Oscilloscope:** The recombined beams produced interference fringes, which were captured by a photodetector and monitored on the oscilloscope for data collection.



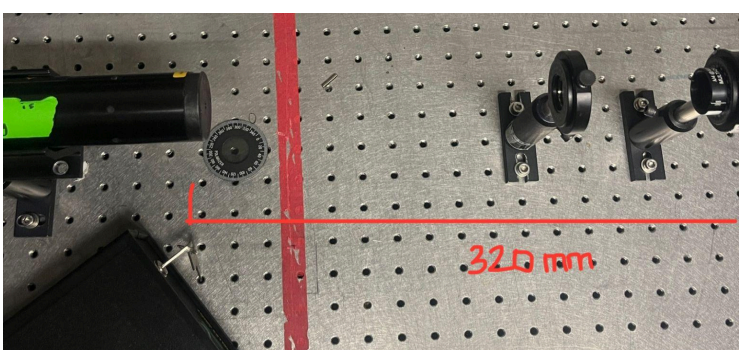
The laser beam was centered and aligned towards the beam splitter. After splitting, one beam traveled toward the fixed mirror while the other reflected off the movable mirror. Both beams recombined at the beam splitter, forming interference fringes at the detector.



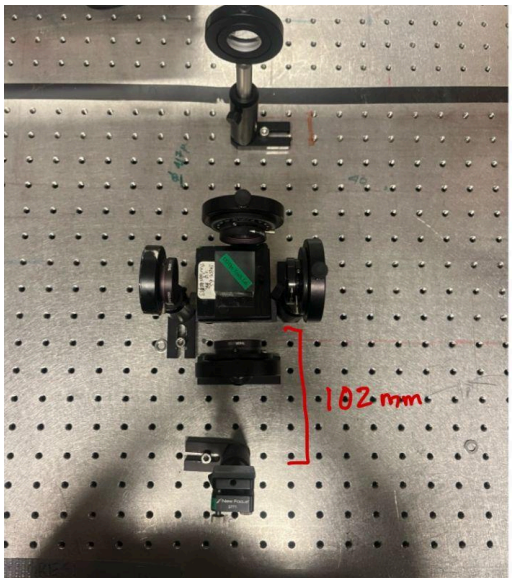
Perfection is key when creating an interferometer. A lot of time was allotted to:

- Align mirrors precisely to ensure the beams overlap.
- Use anti-reflection coated optics to minimize back reflections.
- Adjust the beam splitter height and angle to optimize fringe visibility.

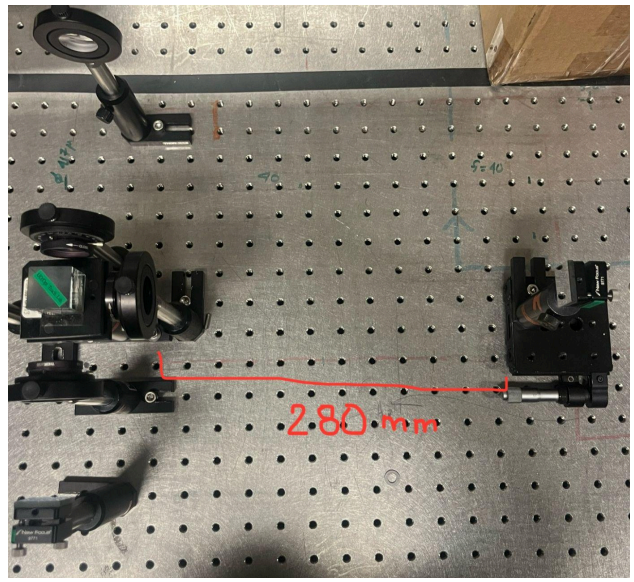
Laser Displacement -
Displacement from beam splitter: 320 mm



Fixed mirror



Movable mirror



Displacement: 102 mm, 280 mm

II. Automation

One of the key goals of this project was to create a way to automatically count interference fringes from the Michelson interferometer data. Traditionally, fringe counting is often done manually, which can be tedious, time-consuming, and prone to human error, especially when dealing with large amounts of data. By automating this process, I was able to improve both the accuracy and efficiency of data analysis.

To automate the fringe counting, I recorded the output signal from the photodetector using an oscilloscope while adjusting the PZT to produce fringes. The oscilloscope data was then exported as CSV files containing voltage versus time information, which reflects the changing intensity of the interference pattern as fringes passed by.

Using Python, I wrote a program that processes these voltage signals. The program applies a Savitzky-Golay filter to smooth out noise in the data while preserving the shape of the fringe signals. After smoothing, the program uses peak detection to identify the locations of the interference fringes.

At first, I decided to focus on the full fringe cycle to get a larger amount of fringes. But after attempting to calculate the wavelength I realized that the full fringe cycle (1 bright + 1 dark) required a different formula than the interferometer wavelength formula.

Therefore, I only counted upward-going fringes (bright) to avoid counting the same fringe multiple times due to noise or oscillation. This approach allowed me to find an accurate count of the total number of fringes corresponding to the displacement applied.

The main benefit of automating fringe counting is that it allows the interferometer to be used for a variety of different measurements:

- **Refractive Index Measurement:** By adding modifications to the setup, automated fringe counting can also be used to measure the refractive index of gases or transparent materials.
- **Thickness Measurements:** Thin film thicknesses can be determined by observing fringe shifts as the optical path length changes.

Code

```python
import sys,pathlib
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import savgol_filter, hilbert, find_peaks

CSV_FILE  = "new.csv"
DISP_UM   = 10.1
WINDOW    = 31                  # SavGol window
POLY      = 3
THR_MIN   = 0.055               # accept peaks ≥ 5.5 %
THR_MAX   = 0.90                # dont allow big spikes
THR_PROM  = 0.027
DIST_S    = 8.0e-6


def count_fringes(csv_path, *, plot=True):
    """
    Count bright Michelson fringes in an oscilloscope CS
V.
    Parameters
    ----------
    csv_path : str | pathlib.Path
        Two-column CSV: column 0 = time [s], column 1 = voltage/in
tensity.
    plot : boo
l       If True, show a diagnostic plot with markers.

    Returns
    -------
    bright_count : int
    """
```

```python
    p = pathlib.Path(csv_path)
    df = pd.read_csv(p)
    t = df.iloc[1:, 0].astype(float).values
    y = df.iloc[1:, 1].astype(float).values

    # smooth and detrend
    y_s = savgol_filter(y, WINDOW, POLY)
    y_d = y_s - np.median(y_s)

    # envelope normalise
    env = np.abs(hilbert(y_d)); env += env.mean()*1e-3
    y_n = y_d / env

    dt = np.mean(np.diff(t))
    dist_pts  = max(1, int(DIST_S / dt))

    max_idx, _ = find_peaks(
        y_n,
        height=(THR_MIN, THR_MAX),
        prominence=THR_PROM,
        distance=dist_pts
    )
    if plot:
        plt.figure(figsize=(8, 3.5))
        plt.plot(t, y_n, lw=1)
        plt.plot(t[max_idx], y_n[max_idx], "r^", label=f"{len(
max_idx)} maxima")
        plt.title(f"{p.name}: {len(max_idx)} bright fringes")
        plt.xlabel("Time (s)"); plt.ylabel("normalised amplitude")
        plt.legend(); plt.tight_layout(); plt.show()

    return len(max_idx)

def wavelength(displacement_m, fringe_count):
    """
    λ = 2·Δs / N
    """
    if fringe_count <= 0:
        raise ValueError("fringe count must be positive")
    return 2.0 * displacement_m / fringe_count

if __name__ == "__main__":
    bright = count_fringes(CSV_FILE, plot=True)
    print(f"Bright peaks = {bright}")

    lam = wavelength(DISP_UM * 1e-6, bright)       # µm → m
    print(f"Estimated λ = {lam*1e9:.2f} nm "
          f"(mirror travel {DISP_UM:.3f} µm)")
```

III. Results

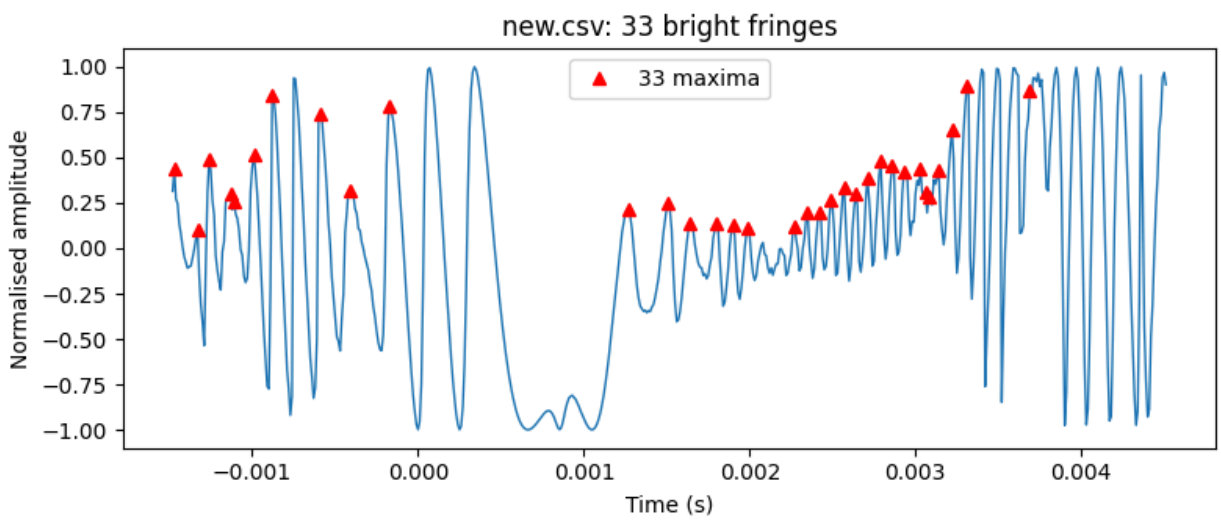
During the experiment, displacement was applied using a PZT actuator because the micrometer stayed fixed and all motion came from the piezo stack. The mirror displacement was set entirely by the applied PZT voltage:

$$\Delta s = k_{PZT} V$$

The total displacement applied to the moving mirror was:

$$\Delta s = 100 nm/V \; x \; 101 \; V = 10.1 \mu m$$

Using the code developed, the number of bright fringes counted during this displacement was 33 fringes.



The laser wavelength was calculated using the standard Michelson interferometer formula for using only bright fringes:

$$\lambda = \frac{2 \cdot \Delta d}{N}$$

Substituting the measured values:

$$\lambda = \frac{2 \cdot 10.1 \mu m}{N} = \frac{2 \cdot 0.01 mm}{33} = 612.12 \text{ nm}$$

This result is close to the expected wavelength of a typical Michelson interferometer. The difference is likely due to uncertainties in displacement calibration, small counting errors, or noise in the signal. There is about a 3.3% error in our measurement. These small variations highlight both the sensitivity of these measurements and the importance of precise alignment and automation.

IV. Conclusion

In this project, I successfully built and operated a Michelson interferometer to explore the principles of optical interference and precision measurement. One of the main goals was to implement an automated approach for fringe counting using data collected from the oscilloscope. Automating the fringe counting process created faster data analysis, improved consistency, and reduced error compared to manual counting. Through these measurements, I was able to calculate the laser's wavelength with reasonable accuracy.

Beyond this experiment, the automated fringe counting system I developed could be applied to a variety of other precision measurements, such as determining refractive indices or thin film thicknesses. Overall, this project demonstrated both the sensitivity and versatility of the Michelson interferometer, while also showing how automation can improve data analysis in physics.